Experiment 3: Configuring a Site-to-Site IPsec VPN Tunnel

Fourth Year – Cybersecurity Engineering Technologies | Cisco Packet Tracer

1. Objective

Build two LANs and configure a secure site-to-site IPsec VPN between routers R1 and R2, then generate and verify encrypted traffic.

2. Key Idea

- A site-to-site VPN securely connects two different local networks across an untrusted network.
- IPsec provides confidentiality, integrity, authentication, and encryption.
- The routers perform encryption/decryption; the PCs send normal IP traffic.

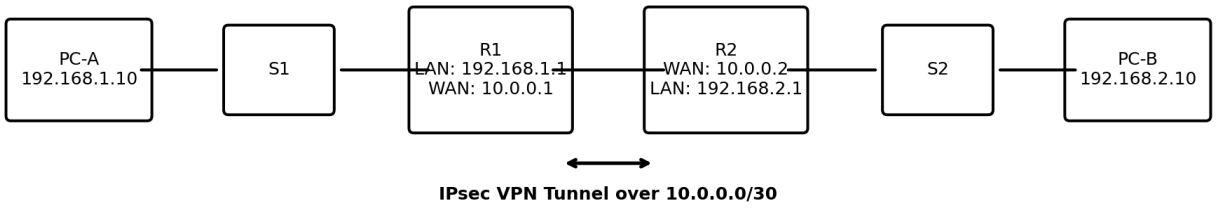


Figure 1. Site-to-Site VPN Topology

3. Addressing Table

Device / Interface	IP Address	Mask	Gateway
PC-A	192.168.1.10	255.255.255.0	192.168.1.1
R1 G0/0	192.168.1.1	255.255.255.0	—
R1 G0/1	10.0.0.1	255.255.255.252	—
R2 G0/1	10.0.0.2	255.255.255.252	—
R2 G0/0	192.168.2.1	255.255.255.0	—
PC-B	192.168.2.10	255.255.255.0	192.168.2.1

4. Build and Test the Basic Network

1. Add 2 × Cisco 2911 routers, 2 × Cisco 2960 switches, and 2 PCs.
2. Connect: PC-A → S1 → R1 → R2 → S2 → PC-B.
3. Configure the PCs using the addressing table.
4. Configure router interfaces and static routes as shown below.

R1 Basic Configuration

```
enable
configure terminal
hostname R1
interface g0/0
```

```
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
interface g0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
exit
ip route 192.168.2.0 255.255.255.0 10.0.0.2
end
```

R2 Basic Configuration

```
enable
configure terminal
hostname R2
interface g0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit
interface g0/1
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.0.0.1
end
```

5. Test Before VPN

Test	Command
PC-A → R1	ping 192.168.1.1
R1 → R2	ping 10.0.0.2
PC-B → R2	ping 192.168.2.1
PC-A → PC-B	ping 192.168.2.10

Important: Do not configure IPsec until the basic ping between the two LANs works.

6. Configure the IPsec VPN

Main steps: interesting traffic ACL → IKE Phase 1 → IPsec transform set → crypto map → apply to WAN interface.

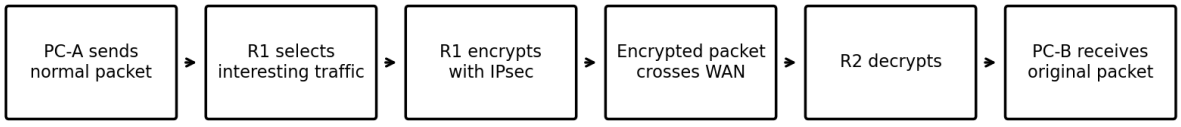


Figure 2. Simplified IPsec VPN Flow

R1 VPN Configuration

```
configure terminal
access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
crypto isakmp policy 10
 encryption aes 256
 hash sha
 authentication pre-share
 group 5
exit
crypto isakmp key vpn123 address 10.0.0.2
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.0.0.2
 set transform-set VPN-SET
 match address 110
exit
interface g0/1
 crypto map VPN-MAP
end
```

R2 VPN Configuration

```
configure terminal
access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
crypto isakmp policy 10
 encryption aes 256
 hash sha
 authentication pre-share
 group 5
exit
crypto isakmp key vpn123 address 10.0.0.1
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.0.0.1
 set transform-set VPN-SET
 match address 110
exit
interface g0/1
 crypto map VPN-MAP
end
```

7. Activate and Verify the Tunnel

1. From PC-A, run: ping 192.168.2.10
2. If the first ping fails, repeat it; the routers may be negotiating the tunnel.
3. On R1, verify IKE with: show crypto isakmp sa

4. Verify IPsec counters with: show crypto ipsec sa

5. Optional: check show crypto map and show access-lists.

Check	Expected Result
PC-A → PC-B ping	Successful
IKE status	Active / QM_IDLE
pkts encaps / encrypt	Greater than 0
pkts decaps / decrypt	Greater than 0

8. Quick Troubleshooting

- No gateway ping: check PC IP, mask, gateway, cable, and router interface no shutdown.
- R1 cannot reach R2: verify 10.0.0.1/30 and 10.0.0.2/30.
- Remote LAN ping fails: verify static routes on both routers.
- VPN does not start: the ACLs must be mirror images and the same pre-shared key must be used.
- Crypto map must be applied on G0/1 of both routers.

9. Student Task / Required Evidence

- Build the topology and configure all IP addresses.
- Configure both static routes and test normal communication.
- Configure the matching IPsec VPN on R1 and R2.
- Generate traffic and verify IKE/IPsec status.
- Save the project as Week3_IPsec_VPN_StudentName.pkt.
- Submit screenshots of topology, successful ping, show crypto isakmp sa, and show crypto ipsec sa.

10. Review Questions

1. What is the purpose of a site-to-site VPN? To securely connect two different LANs through an untrusted network.

2. Which devices encrypt the traffic? R1 and R2.

3. What is interesting traffic? Traffic selected by the crypto ACL for IPsec protection.

4. Which command verifies IKE Phase 1? show crypto isakmp sa

5. Which command shows encrypted/decrypted packet counters? show crypto ipsec sa

Setup the R1 and R2:

```
en
conf t
license boot level securityk9
end
write
reload
```

1. أوامر الراوتر الأول (- (R1النسخة المختصرة جداً

text

```
enable
conf t
int g0/0
 ip add 192.168.1.1 255.255.255.0
 no shut
int g0/1
 ip add 10.0.0.1 255.255.255.252
 no shut
exit

ip route 192.168.2.0 255.255.255.0 10.0.0.2
access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255

crypto isakmp policy 10
 authentication pre-share
exit
crypto isakmp key vpn123 address 10.0.0.2

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.0.0.2
 set transform-set VPN-SET
 match address 110
exit

int g0/1
 crypto map VPN-MAP
end
```

Use code with caution.

2. أوامر الراوتر الثاني (- (R2النسخة المختصرة جداً

text

```
enable
conf t
int g0/0
 ip add 192.168.2.1 255.255.255.0
 no shut
int g0/1
 ip add 10.0.0.2 255.255.255.252
 no shut
exit

ip route 192.168.1.0 255.255.255.0 10.0.0.1
access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255

crypto isakmp policy 10
 authentication pre-share
exit
crypto isakmp key vpn123 address 10.0.0.1

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.0.0.1
 set transform-set VPN-SET
 match address 110
exit

int g0/1
 crypto map VPN-MAP
end
```